

Course- BTech
Course Code- CSET101

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Course Name- Computational Thinking and Programming
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Batch- ALL

Tutorial Assignment: 9

Q1. Determine the output of the following code snippet:

```
Li = ['pear', 'orange', 'apple', 'grapefruit', 'apple', 'pear']  
print(list(set(Li)))  
print(Li.index('apple'))  
print(Li.index('pear'))
```

- a) ['orange', 'pear', 'apple', 'grapefruit']
2
0
- b) ['pear':2, 'orange':1, 'apple':2, 'grapefruit':1]
2
0
- c) 0
- d) None of these

Q2. What does this code print?

```
grocery_list = ['flour', 'cheese', 'carrots']  
for idx, val in enumerate(grocery_list):  
    print("%s: %s" % (idx, val))
```

Q3. Analyze the output of the following code blocks.

```
round1=['chuck norris','bruce lee','sonny chiba']  
round2=round1  
round2.remove('sonny chiba')  
print(round1)  
print(round2)
```

Q4. In the above code, the company was required to keep results of round1 and round2. But by updating results of round2 has also changed List of round1. How to keep results of round2 and round1 separate. Hint: Use shallow copy

Q5. Modify the following code to achieve the specified output:

```
keys = ['r', 'g', 'b', 'o']
vals = [1, 2, 3]
color_dict = dict(zip(keys, vals))
print(color_dict)
```

OUTPUT - {'r': 3, 'g': 2, 'b': 1}

Q6. List the modifications needed for the following program to ensure d and d3 contain identical dictionaries:

```
d1 = {'a':100, 'b':200, 'x':300, 'y':400}
d2 = {'x':500, 'y':600}
d = d1.copy()
d3 = d1
d1['x'] = 800
print(d)
d.update(d1)
print(d)
d.update(d2)
print(d)
print(d3)
```

Q7. Will an error occur on line 5 or line 6? Identify the error.

```
mydict = {}
mydict = {'g':'hello', 's':'how', 't':'this'}
print(mydict['g'])
print(mydict['s'])
print(mydict["t"])
print(mydict['c'])
```

Q8. What is the error in this code snippet:

```
mydict = {'g':'hello', 's':'how', 't':'this'}
print("original dictionary is:"+str(mydict))
update = {'g':17, 's':23, 'this':87}
dict_new = {key:update.get(key,mydict[key]) for key in mydict}
print("updated dictionary is:"+str(dict_new))
print(mydict['this'])
```

Q9. Predict the output of the following code:

```
def check(t):
    return all(i==45 for i in t)

tuple1 = (45,45,45,45)
print(check(tuple1))
```

Q10. Consider a dictionary representing furniture in a room:

```

room={
    'table':'10',
    'chair':'20',
    'beenbags':'40',
    'longchair':'30'}

```

- (a) How we can use get method to get the value of keys? write the code.
- (b) How the quantity of chairs in the room can be updated? Write the code.
- (c) How we can use Pop method in this dictionary. Write the code.
- (d) Can we also use copy method in dictionary? Write the code.

Q11. Create two dictionaries that contain the details of students in two sections of a class. First, input the number of students in a section (dictionary). Then, input the admission number of the students as the keys and their names as the values of the dictionary. Finally, create and display a new dictionary that merges the entries from both sections.

Q12. Execute Union, Intersection, and Difference operations with sets A and B, each containing 100 random integers with a maximum value less than 200.

Q13. What is the output of Python program to concatenate following dictionaries to create anew one.

```

dic1 = {1: 10, 2: 20}
dic2 = {3: 30, 5: 40}
dic3 = {5: 50, 6: 60}
dic4 = {}
for d in (dic1, dic2, dic3):
    dic4.update(d)
print(dic4)

```

Q14. What is the output of the following program.

```

my_dict = {}

if not bool(my_dict):
    print("Dictionary is empty")

```

Q15. Write a Python program to count the values associated with key in a dictionary with format.

```

student = [{'id': 1, 'success': True, 'name': 'Lary'},
            {'id': 2, 'success': False, 'name': 'Rabi'},
            {'id': 3, 'success': True, 'name': 'Alex'}]

```

Q16. What is the output of the following code?

```
dict1 = dict()
for y in enumerate(range(4)):
    dict1[y[0]] = y[1]
    dict1[y[1]+7] = y[0]
print(dict1)
```

Q17.What is the error in the following code?

```
dict1 = {"Amit":100, "Ajay":40}
dict2 = {"Amit":400, "Ajay":40}
print(dict1>dict2)
```

Q18. Split the string sentence into a list of words, then create a dictionary named word_count that contains each word and the number of times it occurs.

Bonus Question:

1. Given a string and a number N, we need to mirror the characters from N-th position up to the length of the string in alphabetical order. In mirror operations, we change 'a' to 'z', 'b' to 'y' and so on.

Input: N=3 Paradox Output: paizwlc